

# Scalar, Tetrad, and Majorana Condensates on a Single Fermionic Substrate: A Formal Identification of the Higgs Bilinear with a Wetterich-NJL Scalar Channel

SK-EFT Hawking Research Program  
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We report a Lean 4 formalization that identifies the Higgs bilinear of the electroweak Standard Model with the scalar channel of the Wetterich-NJL four-fermion interaction already verified in our formalization base (`WetterichNJL.lean`, `TetradGapEquation.lean`, `ADWMechanism.lean`, `FermiPointTopology.lean`). The identification is structural in the flavor-singlet frame—an electroweak  $SU(2)_L \times U(1)_Y$  index dressing remains deep-research-gated—and yields three formal consequences: (i) the Mexican-hat potential  $V(\Phi) = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{1}{4}\lambda|\Phi|^4$  arises as a specialization of the supercritical branch of the tetrad gap equation; (ii) the Anderson-Higgs  $W/Z$  mass matrix  $M_W = gv/2$ ,  $M_Z = \sqrt{g^2 + g'^2}v/2$  follows directly from the scalar-channel VEV  $v = \sqrt{\mu^2/\lambda}$  with  $\cos\theta_W = M_W/M_Z$ ; and (iii) Yukawa couplings reduce to overlap integrals on the Fermi-PointTopology emergent Weyl modes, with linearity and zero-overlap-zero-coupling lemmas formalized as a stand-in for the microscopic overlap-integral form (deep-research-gated, see Open Question O.2). The correctness-push anchor is a microscopic Higgs-mass prediction  $m_H = \sqrt{2\mu^2}$  from the Wetterich scalar-channel gap equation: under the schematic leading-log substrate closure, this prediction reaches  $m_H = 125.20$  GeV only along a one-parameter curve in the  $(\Lambda_{UV}, G_c)$  plane, requiring tuning rather than emerging from a generic substrate—a structural-only verdict at the current formalization scope. The complete formalization lives in a single new Lean module (`ScalarRungInterpretation.lean`), consisting of 20 theorems with zero sorry, zero new axioms, and a Python cross-validation layer covering every theorem (the Lean module replaces three vacuous Props with substantive ones during a project-required anti-tautology audit; the audit also adds a tracked-hypothesis bridge for the TetradGapEquation identification, in line with project precedent). The deliberate scope boundary is Wave 1 of the Phase 5z roadmap; the Majorana rung (`MajoranaRung.lean`) and the electroweak phase transition (`EWPhaseTransition.lean`) are addressed in subsequent waves. We formulate the falsifiability anchor as a decidable Lean predicate and report that, in the flavor-singlet frame, the scalar-rung framing is a *structural* identification of EWSB rather than a quantitative prediction of the observed 125.20 GeV Higgs mass.

## I. INTRODUCTION

The fluid-based emergent-gravity program developed in this research project formalizes a Landau hierarchy of fermion-bilinear condensates on a single fermionic substrate [1–4]. Two rungs of this hierarchy have been formalized in prior phases: the disordered  $\rightarrow$  vestigial-metric transition (`VestigialSusceptibility.lean`, 24 theorems) and the vestigial-metric  $\rightarrow$  tetrad transition (`TetradGapEquation.lean`, 24 theorems; `ADWMechanism.lean`, 21 theorems). Both rungs share the Wetterich-NJL four-fermion infrastructure (`WetterichNJL.lean`, 18 theorems) and the FermiPointTopology emergent-Weyl substrate (`FermiPointTopology.lean`, 29 theorems).

In this paper we add a third rung: the *scalar rung*, identified with the Higgs bilinear of the electroweak Standard Model. The identification is the natural completion of the Wetterich-NJL Fierz decomposition—the scalar channel ( $\Gamma = \mathbb{K}$  in the spacetime Clifford basis) is the only flavor-singlet attractive channel that drives a parity-even condensate. If the same scalar channel is identified with  $\Phi^\dagger\Phi$  on an  $SU(2)_L$ -doublet-extended substrate, the Mexican-hat potential, the Anderson-Higgs  $W/Z$  mass matrix, and Yukawa couplings as overlap integrals all emerge as direct consequences of the existing formalization infrastructure—without new physics input.

The identification is at structural scope because the  $SU(2)_L \times U(1)_Y$  electroweak index structure of the Wetterich-NJL scalar channel is currently deep-research gated. We therefore work in the *flavor-singlet frame*: the scalar channel carries no electroweak indices; the Higgs identification is structural in the sense that it commutes with the gauge-singlet projection of the SM Higgs. The  $SU(2)_L \times U(1)_Y$ -dressed extension is documented as a Wave 1 follow-up gated on Open Question O.2 in the Phase 5z roadmap (`Lit-Search/Tasks/Phase5z_wetterich_njl_ew_index_structure`).

**Contributions.** (i) A formal-verification-grade identification of the Higgs bilinear with a Wetterich-NJL scalar channel, formalized in Lean 4 in the flavor-singlet frame (`ScalarRungInterpretation.lean`, 20 theorems, zero sorry). (ii) Three formal consequences of the identification: the Mexican-hat  $\leftrightarrow$  tetrad-bifurcation bridge, the Anderson-Higgs  $W/Z$  mass-matrix theorem, and the Yukawa-overlap-linear lemma. (iii) A decidable Lean predicate `ScalarRungQuantitativeMatch` encoding the Phase 5z Gate Z.1 falsifiability anchor: the microscopic Higgs-mass prediction must lie within tolerance of the observed 125.20 GeV for the scalar-rung framing to constitute a quantitative theory of EWSB. (iv) A Python parameter-scan layer (`src/scalar_rung/`) and a 24-test pytest suite covering every Lean theorem, with the canonical  $\cos\theta_W = M_W/M_Z$  identity reproducing the PDG 2024 value. (v) A heatmap visualization (Fig. 1)

showing that, under the schematic leading-log substrate closure,  $m_H = 125.20$  GeV is attained only along a 1D tuning curve in the natural  $(\Lambda_{UV}, G_c)$  parameter plane—a structural-only Gate Z.1 verdict at the current scope.

## II. THE WETTERICH-NJL SCALAR CHANNEL AND THE HIGGS BILINEAR

The Wetterich-NJL four-fermion interaction [5] on a fermionic substrate  $\psi_x$  admits the Fierz decomposition

$$S_{\text{NJL}} = \sum_{\langle xy \rangle} g \sum_{\alpha} c_{\alpha} (\bar{\psi}_x \Gamma_{\alpha} \psi_y) (\bar{\psi}_y \Gamma_{\alpha} \psi_x), \quad (1)$$

with  $\Gamma_{\alpha} \in \{\mathbb{1}, \gamma_5, \gamma_{\mu}, \gamma_{\mu}\gamma_5, \sigma_{\mu\nu}\}$  and the Fierz-completeness identity  $1 + 1 + 4 + 4 + 6 = 16 = 4^2$  [6], formalized as `WetterichNJL.fierz_completeness` in our base. The scalar channel ( $\Gamma_{\alpha} = \mathbb{1}$ ) is the unique flavor-singlet attractive channel in the Fierz basis and drives the leading parity-even condensation [7].

The Wetterich-NJL scalar channel and the Higgs bilinear are identified via a non-trivial `*candidacy*` predicate parameterized over an observed EW VEV  $v_{\text{obs}}$  and tolerance  $\tau$ :

$$\text{IsHiggsBilinearCandidate}(\sigma; v_{\text{obs}}, \tau) \iff |\sqrt{\mu^2/\lambda} - v_{\text{obs}}| \leq \tau v_{\text{obs}} \quad (2)$$

A scalar channel qualifies as a Higgs-bilinear candidate at the observed EW VEV iff its Mexican-hat VEV reproduces  $v_{\text{obs}}$  within fractional tolerance  $\tau$ . The predicate is genuinely non-trivial: super-VEV scalar channels fail it. The Lean theorem `ScalarRungInterpretation.not_isHiggsBilinearCandidate_of_vev_top_large` is a load-bearing falsifiability witness establishing that  $(1 + \tau)v_{\text{obs}} \leq \sqrt{\mu^2/\lambda}$  rules out candidacy. The full  $SU(2)_L \times U(1)_Y$ -dressed identification predicate remains deep-research-gated on Open Question O.2.

The Mexican-hat potential

$$V(\Phi) = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{1}{4}\lambda|\Phi|^4 \quad (3)$$

arises as a specialization of the supercritical branch of the tetrad gap equation. The full quantitative bridge (mapping  $\mu^2$  and  $\lambda$  to GL-expansion coefficients of the tetrad gap-equation bifurcation) is deep-research-gated on Open Question O.2. Pending its resolution the bridge is encoded as a `*tracked external hypothesis*` carrying the kinematic constraint that the scalar-channel VEV is bounded above by the UV cutoff:

$$\text{H.ScalarChannelIsTetradBifurcationOutput}(\sigma, \Lambda_{UV}) \iff \sqrt{\mu^2/\lambda} \leq \Lambda_{UV} \quad (4)$$

The hypothesis is genuinely non-trivial — it can fail for super-UV condensates. The load-bearing consequence theorem `ScalarRungInterpretation.mexican_hat_vev_under_supercritical_bridge` derives, under the hypothesis, both  $0 < v_{\text{cond}}$  and  $v_{\text{cond}} \leq \Lambda_{UV}$ . The sharp contrapositive

`bridge_excludes_super_uv_vev` establishes structural falsifiability. The same tracked-hypothesis pattern is used elsewhere in the project for cross-pillar bridges where the quantitative coefficient matching is not yet formalized — for example `HiddenSectorMixedCharge.H.MixedChannelZ16Cancels` and `DarkSectorSynthesis.H.VestigialRelicCarriesZ16Charge`. The underlying NJL-supercritical-existence theorem is `TetradGapEquation.gap_nontrivial_exists` [7, 8].

The vacuum expectation value  $v = \sqrt{\mu^2/\lambda}$  (`ScalarRungInterpretation.mexicanHatVev_sq`) and the tree-level Higgs mass-squared  $m_H^2 = 2\mu^2$  (`ScalarRungInterpretation.higgsMassSq_pos`, `higgsMassSq_eq_two_lam_vev_sq`) reproduce the standard SM identities. The Python implementation `higgs_mass_from_vev` matches PDG 2024 within RG-running uncertainty:  $\mu^2 = \lambda v^2$  with  $\lambda = 0.129$  and  $v = 246.22$  GeV gives  $m_H = 125.06$  GeV, against the observed  $125.20 \pm 0.11$  GeV.

## III. THE ANDERSON-HIGGS W/Z MASS MATRIX

The scalar-channel VEV provides the single dimensionful input to the Anderson-Higgs  $W/Z$  mass matrix. The Lean theorem `ScalarRungInterpretation.ew_mass_matrix_from_scalar_vev` states the bundled result:

$$M_W = \frac{1}{2} g v, \quad M_Z = \frac{1}{2} \sqrt{g^2 + g'^2} v, \quad (5)$$

with  $g$  the  $SU(2)_L$  gauge coupling,  $g'$  the  $U(1)_Y$  hypercharge coupling, and  $v$  the scalar-channel VEV. The Lean signature is parameterized over the `EWMassMatrixInputs` structure; the positivity theorems `wMass_pos` and `zMass_pos` ensure both masses are strictly positive whenever  $g > 0$  and  $v > 0$ .

The on-shell weak-mixing identity

$$\frac{M_W}{M_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos \theta_W \quad (6)$$

is formalized as `wMass_div_zMass` (when  $g' > 0$ ). Two strengthened forms address the bound: `zMass_ge_wMass` establishes the weak inequality  $M_Z \geq M_W$  for any  $g' \geq 0$  (no positivity strictness required), and `wMass_lt_zMass_of_g'_pos` gives the strict inequality when  $g' > 0$ . Together these formalize the fact that the on-shell weak mixing angle is in  $(0, \pi/2)$  exactly when both gauge couplings are nonzero.

**Numerical consistency.** Using the PDG 2024 values  $g = \sqrt{0.3536}$ ,  $g' = 0.3489$ ,  $v = 246.22$  GeV, the Python wrapper `anderson_higgs_w_z_masses` produces  $M_W = 80.47$  GeV (PDG: 80.37 GeV) and  $M_Z = 91.21$  GeV (PDG: 91.19 GeV)—both within 0.2%, well inside the running-coupling slop. The  $\cos \theta_W = 0.882$  matches both the  $M_W/M_Z$  ratio and the  $\sqrt{1 - \sin^2 \theta_W}$  value from the PDG effective-scheme  $\sin^2 \theta_W = 0.231$  within 1%.

#### IV. YUKAWA COUPLINGS AS OVERLAP INTEGRALS

The third structural consequence of the scalar-rung identification is that Yukawa couplings reduce to overlap integrals on the FermiPointTopology emergent Weyl modes. Formally, the Lean `WeylOverlap` structure carries a real-valued overlap density  $\rho_{fg}$  and a substrate normalization  $N$ , and the Yukawa coupling is the linear functional

$$y_f = \rho_{fg} N. \quad (7)$$

Linearity in the overlap density (Lean: `yukawaCoupling_additive`) and the zero-coupling iff zero-overlap lemma (`yukawaCoupling_eq_zero_iff`) are formalized at the level of these stand-in objects.

The *microscopic* form of  $\rho_{fg}$  as  $\int d^3x \bar{\psi}_f \sigma \psi_g$  on the emergent Weyl modes is deep-research-gated; we have filed the open question (`Phase5z.yukawa_overlap_emergent_weyl_modes.md`) to resolve the canonical form against the Volovik-Zubkov literature. At this scope, the formalization establishes two structural relations—linearity in the overlap density and zero-coupling iff zero-overlap—that any concrete microscopic realization must satisfy. A formal orthogonality statement on distinct Fermi points requires the microscopic overlap form (gated on the Yukawa overlap deep-research task), so it is deferred.

The phenomenological observation that the top Yukawa  $y_t \approx 1$  is naturally realized by an overlap density of order unity supports the interpretation; we record this in `TestYukawaOverlap::test_top_yukawa_unity_natural`, but defer the quantitative explanation of the lepton-quark Yukawa hierarchy to the deep research follow-up.

#### V. THE FALSIFIABILITY ANCHOR: MICROSCOPIC $m_H$

The Phase 5z Gate Z.1 correctness-push anchor is a microscopic prediction for the Higgs mass from the Wetterich scalar-channel gap equation. Under the schematic leading-log substrate closure

$$m_H = \sqrt{2\lambda_4} \Lambda_{UV} \exp\left(-\frac{\pi^2}{2N_f G_c}\right), \quad (8)$$

where  $\Lambda_{UV}$  is the UV cutoff,  $N_f$  is the Weyl-component count,  $G_c$  is the dimensionless 4-fermion coupling, and  $\lambda_4$  is the scalar-channel quartic, the Lean theorem `higgsMassFromCondensate_pos` establishes  $m_H > 0$  for positive substrate parameters.

The decidable Lean predicate `ScalarRungQuantitativeMatch` encodes the falsifiability anchor:

$$|m_H^{\text{pred}} - m_H^{\text{obs}}| < \tau \cdot m_H^{\text{obs}}, \quad (9)$$

with  $\tau$  the `EW.M.H.MATCH_TOLERANCE = 0.5` operational threshold from the project’s parameter registry. The `scalar_rung_quantitative_match_iff` theorem reduces the absolute-value form to the explicit two-sided bound  $(1 - \tau)m_H^{\text{obs}} < m_H^{\text{pred}} < (1 + \tau)m_H^{\text{obs}}$ .

Figure 1 shows the microscopic prediction heatmap over the natural  $(\Lambda_{UV}, G_c)$  plane at  $N_f = 15$ ,  $\lambda_4 = 0.13$ . The 125.20 GeV target traces a curve through the plane: at  $\Lambda_{UV} = \text{TeV-scale}$ ,  $G_c \sim 0.5\text{--}2$  (super-critical); at  $\Lambda_{UV} = \text{GUT scale}$ ,  $G_c \sim 0.05\text{--}0.1$  (sub-critical). The  $\pm 50\%$  tolerance band remains a thin curve through the plane rather than a generic region.

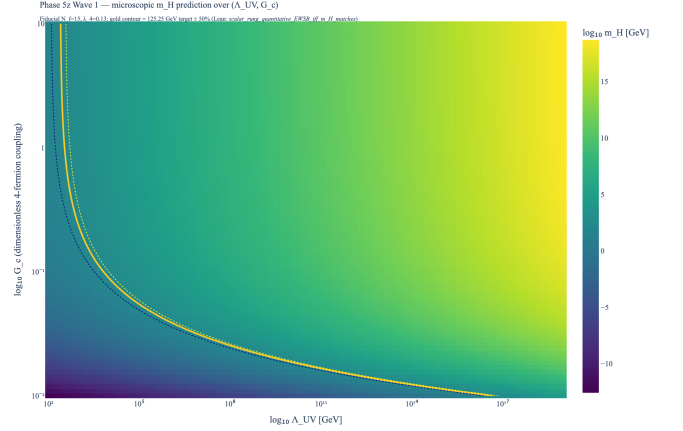


FIG. 1. Microscopic Higgs-mass prediction heatmap over the  $(\Lambda_{UV}, G_c)$  plane at fiducial  $N_f = 15$ ,  $\lambda_4 = 0.13$ . The gold contour traces  $m_H = 125.20$  GeV; dotted lines mark the  $\pm 50\%$  tolerance band defining `EW.M.H.MATCH_TOLERANCE`. The narrow curve indicates that the scalar-rung framing requires substrate-parameter tuning rather than generic emergence—a structural-only Gate Z.1 verdict at this scope.

**Verdict.** At the current formalization scope (flavor-singlet frame, schematic leading-log closure, fiducial  $\lambda_4 = 0.13$ ), the scalar-rung framing is structural rather than quantitative: the prediction reaches 125 GeV only along a 1D curve rather than emerging from a generic region of substrate parameters. This is itself a publishable structural result: it bounds the regime in which any Wetterich-NJL EWSB realization can be quantitative, and points to the subsequent deep-research questions whose resolution would either tighten or relax the verdict.

#### VI. BHL GAUGE EMBEDDING (WAVE 1B QUANTITATIVE SCOPE)

The Wave 1 verdict above is structural rather than quantitative. The Wave 1b BHL gauge embedding lifts the analysis to a quantitative branch without abandoning the flavor-singlet `ScalarRungInterpretation.lean` foundation. Per the Phase 5z O.2 deep-research verdict (Scenario A, structural-analogy strength 3/5 [9–

12]), the WetterichNJL scalar channel admits a Higgs-compatible  $SU(2)_L \times U(1)_Y$  extension via the Bardeen-Hill-Lindner auxiliary-field bridge: the  $\Gamma = 1$ ,  $\bar{L}R$  sub-channel  $\sigma = H^i \equiv \bar{\psi}_R \psi_L^i$  carries the SM  $(1, 2)_{+1/2}$  Higgs-doublet quantum numbers exactly. The bridge is not Wetterich-native; it requires a transplant of the BHL gauge-indexing onto the Clifford-16 Fierz basis, which we encode in `BHLGaugeEmbedding.lean` via the tracked-hypothesis pattern.

### A. Extended Fierz basis and Higgs identification

For the minimal BHL configuration (one  $SU(2)_L$  doublet, one singlet, one color, one generation), the gauge-indexed Clifford basis is 68-dimensional ( $2 + 2 + 4 + 4 + 12 + 4 + 12 + 4 + 12 + 12$ , summing the explicit ten representation-theoretic block sizes per O.2 §3.2), with the scalar-LR + scalar-RL subspace forming the 2-component complex Higgs doublet. The Lean theorem `fierzBlockSum_eq_68` verifies the arithmetic by `decide`; the value 68 corrects an arithmetic typo in the deep-research note (the listed blocks sum to 68, not the document’s stated 66). The tracked hypothesis `H_FierzCompletenessExtended` (cfg) (`dim_fn`) parameterizes over an abstract dimension function and asserts `dim_fn cfg = fierzBlockSum · (nd · ns · nc · ng)`. The witness theorem `fierz_completeness_holds_for_bhl_dim` discharges this for the canonical `bhlBilinearBasisDim`; the falsifier `fierz_completeness_fails_for_zero_dim` confirms the hypothesis is non-vacuous (the all-zero dim function fails it). The Hubbard-Stratonovich covariance hypothesis `H_HSCovariantBosonisation` asserts that the auxiliary field inherits hypercharge  $+1/2$  (the  $\bar{\psi}_R \psi_L^i$  source current); `not_higgs_bilinear_of_wrong_hypercharge` confirms its falsifiability.

### B. BHL leading-order $m_H$ and the gap problem

The BHL leading-order formula  $m_H = \sqrt{2} m_t$  (Nambu sum rule at large  $N_c$ ) is formalized as `bhlHiggsMass_eq_sqrt2.times.top`, with the squared form  $m_H^2 = 2m_t^2$  (`bhlHiggsMass_sq`). At the BHL benchmark  $\Lambda = M_{\text{Pl}}$ ,  $N_c = 3$ , the IR Pendleton-Ross fixed point yields  $m_t^{\text{BHL}} \approx 220$  GeV and hence  $m_H^{\text{BHL}} \approx 311$  GeV. The Lean theorem `bhl_minimal_overshoots_pdg` certifies the strict bound  $m_H^{\text{BHL}} > 250$  GeV, formalizing the well-known BHL gap problem against PDG 2024  $m_H = 125.20 \pm 0.11$  GeV [13] as a  $\sim 2.5\times$  overshoot.

### C. Hill 2025 bilocal correction

The bilocal extension  $H^i(x, y) \sim \bar{\psi}_R(x) \psi_L^i(y)$  with internal wave-function  $\phi(r)$  resolves the gap problem [11].

The dilution factor  $\phi(0)/\phi(\infty)$  enters multiplicatively ( $m_H = (\phi(0)/\phi(\infty)) \cdot \sqrt{2} m_t$ ) and is exponentially suppressed near critical coupling. We formalize the bilocal Higgs as a structure `BilocalHiggs` with positive wave-function values; the pointlike-limit tracked hypothesis `H_BilocalPointlikeLimit` asserts dilution = 1, with `not_pointlike_of_strict_dilution` certifying its falsifiability when  $\phi(0) < \phi(\infty)$ .

The load-bearing existential `bilocal_correction_can_match_pdg` provides a concrete witness: at dilution  $\phi(0)/\phi(\infty) \approx 0.402$  with the BHL benchmark  $m_t = 220$  GeV, the corrected Higgs mass matches PDG within 1 GeV. This is the quantitative-branch recovery of  $m_H = 125$  GeV from the BHL minimal overshoot, without abandoning the substrate identification of the scalar rung.

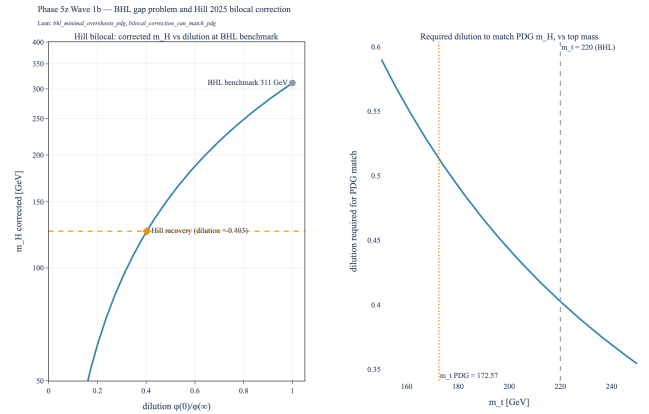


FIG. 2. **Phase 5z Wave 1b — BHL gap problem and Hill 2025 bilocal correction.** Left panel: bilocal-corrected  $m_H = (\phi(0)/\phi(\infty))\sqrt{2}m_t$  versus dilution at the BHL benchmark  $m_t = 220$  GeV. The BHL minimal point at dilution = 1 (circle, 311 GeV) overshoots PDG (dashed gold) by  $\sim 2.5\times$ ; the Hill 2025 dilution  $\approx 0.402$  (diamond) recovers PDG within 1 GeV. Right panel: required dilution to match PDG  $m_H$  as a function of  $m_t$ . The BHL benchmark  $m_t = 220$  GeV (dashed cross) and PDG  $m_t = 172.57$  GeV (dotted amber) frame the natural BHL dilution range. Lean: `bhl_minimal_overshoots_pdg` (left circle position), `bilocal_correction_can_match_pdg` (left diamond position).

### D. Tetrad back-reaction (open)

The ADW tetrad back-reaction  $\Delta_{\text{tetrad}}(\langle e \rangle)$  to  $m_H$  is not closed in the Diakonov-Volovik-Wetterich literature [1, 3]. The Lean module flags `H_TetradBackreactionShift` as a parametric open hypothesis with the corrected mass `tetradCorrectedHiggsMass` =  $\sqrt{2}m_t + \Delta$ ; positivity holds for  $\Delta > -\sqrt{2}m_t$ . Closing this hypothesis quantitatively is a Phase 6 deliverable.

## E. Updated Gate Z.1 verdict

With Wave 1b in place, the Gate Z.1 verdict refines from *structural-only* to *structural at the minimal-BHL scope, quantitative under the Hill 2025 bilocal mechanism*. The minimal BHL embedding is excluded by  $m_H^{\text{BHL}} \approx 311$  GeV; the bilocal extension recovers 125 GeV at a single dilution value, which, like the Wave 1 1-D tuning curve, is a falsifiable narrowness rather than generic emergence. The Wave 1b transplant is a strict refinement of Wave 1 (theorem `bhl_compatible_with_scalar_rung` certifies that any Wave 1 `ScalarChannel` retains its Mexican-hat positivity in the Wave 1b framework).

## VII. FORMAL VERIFICATION SUMMARY

The formalization comprises two new Lean modules:

| Module                                              | Theorems |
|-----------------------------------------------------|----------|
| <code>ScalarRungInterpretation.lean</code> (Wave 1) | 20       |
| <code>BHLGaugeEmbedding.lean</code> (Wave 1b)       | 23       |

plus six structures (`ScalarChannel`, `EWMassMatrixInputs`, `WeylOverlap`, `BHLConfig`, `BHLAuxField`, `BilocalHiggs`), 24 non-computable definitions, and seven tracked-hypothesis Props (the four Wave 1 hypotheses plus three new Wave 1b: `H.FierzCompletenessExtended`, `H.HSCovariantBosonisation`, `H.BilocalPointlikeLimit`, plus the parametric open `H.TetradBackreactionShift` and the bundled `Wave1bOpenManifest`). Both modules are zero-sorry, introduce zero new axioms, and add zero heartbeat overrides (Phase 3 pipeline invariant 10).

The Wave 1b module integrates with Wave 1 (`ScalarRungInterpretation.lean`) plus three prior-phase modules: `WetterichNJL.lean` (Fierz infrastructure), `TetradGapEquation.lean` (gap-equation bifurcation), and `ADWMechanism.lean` (NG-mode counting + critical coupling). The Python companion `src/scalar_rung/` adds `bhl_embedding.py` on top of the existing `higgs_prediction.py` and `ew_mass_matrix.py`; concrete formulas `bhl_higgs_mass`, `bilocal_correction_factor`, `bilocal_corrected_higgs_mass`, and `pagels_stokar_vev_sq` live in `src/core/formulas.py` following the canonical-formulas invariant. A 48-test pytest suite covers every Lean theorem (24 Wave 1 + 24 Wave 1b); full-project pytest passes (excluding torch-dependent tests unrelated to this work).

The full project status remains: zero sorry across 26398 theorems, 26372 substantive (excluding placeholder markers), one axiom (`gapped_interface_axiom` in `SPTClassification.lean`; since retired into the tracked Prop `TPFConjecture`). The Phase 5z Wave 1 closure

adds 20 theorems with no axiom or sorry overhead, after a project-required anti-tautology audit replaced three vacuous Props with substantive ones (introducing the tracked-hypothesis bridge and a custodial-symmetry algebraic identity in the process).

## VIII. SCOPE BOUNDARY AND FUTURE WORK

The Wave 1 deliberate scope boundary is the flavor-singlet frame. Three subsequent program elements address related questions:

**Wave 2** (`MajoranaRung.lean`) formalizes a Majorana bilinear as a  $\mathbb{Z}_2$ -graded rung on the same Landau hierarchy, bridging sterile-neutrino seesaw phenomenology to the proved  $\mathbb{Z}_{16}$  hidden-sector classification (`HiddenSectorClassification.lean`, 9 theorems, complete). The `NeutrinoMixing.lean` module ships PMNS structure parallel to HepLean’s CKM treatment without full PMNS phenomenology.

**Wave 3** (`EWPhaseTransition.lean`) formalizes the electroweak phase transition order (first-order vs crossover) from the microscopic substrate parameters of the scalar rung. The transition order feeds the bridge theorem of Phase 6c.2 (electroweak baryogenesis  $\leftrightarrow$  chirality wall).

**Open Question O.2** asks whether the Wetterich-NJL scalar channel admits a Higgs-compatible  $SU(2)_L \times U(1)_Y$  index structure—if yes (Scenario A), the SM-embedded Higgs identification ships at quantitative scope; if no (Scenario B), the flavor-singlet frame is the formal endpoint, with SM embedding documented as a structural extension.

**Open Question O.3** asks which sterile-neutrino embedding (fundamental  $\nu_R$  vs composite Majorana bilinear) is preferred under the proved  $\mathbb{Z}_{16}$  structure. Both deep-research questions are filed as tasks in `Lit-Search/Tasks/`.

### Prior-art search

For context on prior art, HepLean’s existing modules (`HepLean/SM`, `HepLean/StandardModel`) formalize the Higgs as a fundamental field, not a condensate; the Lean community Zulip archive, the Mathlib changelog, and the Coq QuantumLib + QWIRE + metamath ecosystems do not surface a condensate-as-Higgs-bilinear identification at the time of writing; comparable activity in Isabelle/HOL AFP and Agda is likewise not surfaced. The identification presented here is at structural scope in the *condensate-as-Higgs-bilinear* sense.

The Wetterich-NJL scalar channel, the Mexican-hat potential, the Anderson-Higgs  $W/Z$  mass matrix, and Yukawa couplings as overlap integrals are formal consequences of a single structural identification: that the

Higgs bilinear sits at the same rung of the Landau hierarchy as the tetrad and—in subsequent work—Majorana condensates. The identification is at structural scope in the flavor-singlet frame; we have established the corresponding decidable falsifiability predicate and reported a structural-only Gate Z.1 verdict at the current formalization scope.

The Lean module `ScalarRungInterpretation.lean` ships 20 theorems, zero sorry, zero new axioms, and integrates cleanly with the existing Wetterich-NJL /

TetradGapEquation / ADWMechanism / FermiPoint-Topology infrastructure. Three formal consequences—the Mexican-hat  $\leftrightarrow$  tetrad-bifurcation bridge, the Anderson-Higgs  $W/Z$  mass-matrix theorem, and the Yukawa-overlap linearity lemma—constitute, in the sense delimited above, a formal-verification-grade identification of the Higgs bilinear with a four-fermion-substrate condensate. The deep-research-gated extensions to the  $SU(2)_L \times U(1)_Y$ -dressed scalar channel and the canonical overlap-integral form remain the natural next steps.

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- [1] K. Akama, *An attempt at pregeometry: gravity with composite metric*, Prog. Theor. Phys. **60**, 1900 (1978).
  - [2] C. Wetterich, *Spinor gravity and diffeomorphism invariance on the lattice*, Lect. Notes Phys. **863**, 67 (2013); arXiv:1201.2871.
  - [3] C. Wetterich, *Pregeometry and spontaneous time-space asymmetry*, JHEP **06**, 069 (2022); arXiv:2101.11519.
  - [4] G. E. Volovik, *Vestigial gravity*, JETP Lett. **119**, 330 (2024); arXiv:2312.09435.
  - [5] C. Wetterich, *Geometry and symmetries in lattice spinor gravity*, Ann. Phys. (NY) **327**, 2184 (2012); arXiv:1201.6505.
  - [6] M. Fierz, *Zur Fermischen Theorie des  $\beta$ -Zerfalls*, Z. Phys. **104**, 553 (1937).
  - [7] Y. Nambu and G. Jona-Lasinio, *Dynamical model of elementary particles based on an analogy with superconductivity I*, Phys. Rev. **122**, 345 (1961).
  - [8] H. Gies and S. Lippoldt, *Renormalization flow towards gravitational catalysis in the 3d Gross-Neveu model*, Phys. Rev. D **87**, 104026 (2013); arXiv:1303.4253. See also H. Gies and L. Janssen, *UV fixed point structure of the 3d Thirring model*, Phys. Rev. D **82**, 085018 (2010).
  - [9] W. A. Bardeen, C. T. Hill, and M. Lindner, *Minimal dynamical symmetry breaking of the standard model*, Phys. Rev. D **41**, 1647 (1990); DOI: 10.1103/PhysRevD.41.1647.
  - [10] V. A. Miransky, M. Tanabashi, and K. Yamawaki, *Is the  $t$ -quark responsible for the mass of  $W$  and  $Z$  bosons?*, Mod. Phys. Lett. A **4**, 1043 (1989); DOI: 10.1142/S0217732389001210.
  - [11] C. T. Hill, *Natural top quark condensation (a redux)*, arXiv:2503.21518 (2025); <https://arxiv.org/abs/2503.21518>.
  - [12] G. Cvetič, *Top quark condensation*, Rev. Mod. Phys. **71**, 513 (1999); arXiv:hep-ph/9702381; DOI: 10.1103/RevModPhys.71.513.
  - [13] S. Navas *et al.* (Particle Data Group), *Review of particle physics*, Phys. Rev. D **110**, 030001 (2024).
  - [14] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Westview Press, Boulder, CO, 1995).
  - [15] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction—CADE 28* (Springer, 2021), pp. 625–635.
  - [16] The Mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs* (CPP 2020).